



Introduction to Building Materials - Stones

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Stones - Definition

- ❖ Stones are naturally occurring, hard, compact, and solid masses of minerals that form a part of the Earth's crust.
- ❖ In geology, stones are referred to as rocks. Rocks occur in a wide variety of types and compositions. Only those rocks that possess suitable physical, mechanical, and durability properties are used as building stones.
- ❖ Since prehistoric days it has been used for constructing different components of buildings like foundations, walls, lintels, floors, roof, bridges, wiers, dams etc.





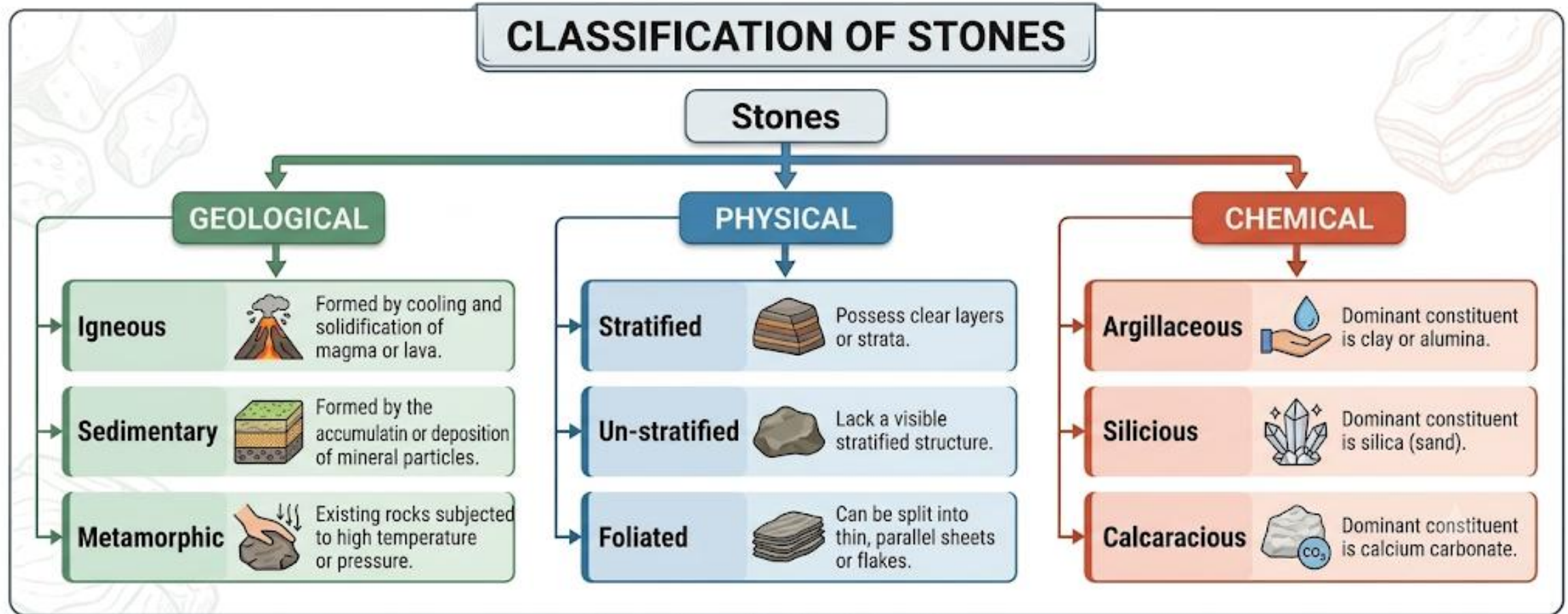
Stones - Introduction

- ❖ Stones have been used for construction since the time immortal e.g. The Pyramids of Egypt.
- ❖ Most of the ancient temples and forts of India were built with stones. The Taj and the Victoria Memorial (Calcutta) were built in white marble had from Rajasthan. Red Fort and Jama Masjid (Delhi) were built with red sand stone from Agra. Parliament House, Rashtrapati Bhawan and the Central Secretariat (Delhi) were built with red and grey sand stones from Rajasthan.





Stones - Classification



Educational Infographic | Classification of Stones



Igneous Rocks

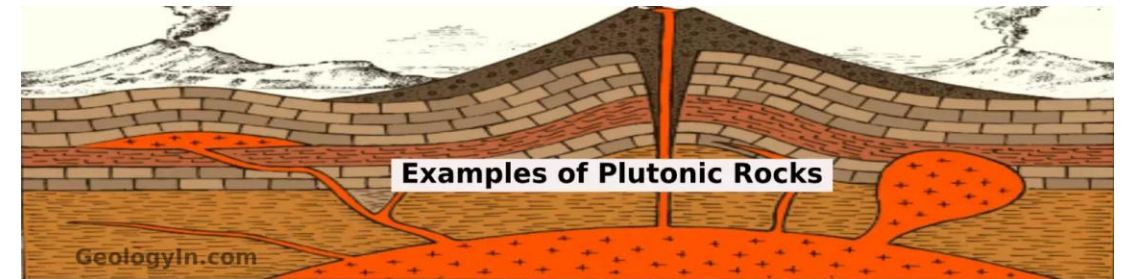
- ❖ Molten rock materials found below the earth's crust are known as magma. During volcanic eruption, this magma, under very temperature and pressure, and varieties of complex phenomena occurring below earth's crust beyond the comprehension of human being, comes out to the surface.
- ❖ The rocks formed due to cooling and consolidation of molten magma on the surface is known as igneous rock.
- ❖ They are the most abundant rocks in the Earth's crust (by volume). Igneous rocks are further classified by geologists into three subclasses based on their depth of formation.





Igneous Rocks - Plutonic

- ❖ These rocks are formed at great depths below the Earth's surface by the slow cooling and solidification of magma.
- ❖ Over a long period, they become exposed at the surface due to the erosion of the overlying rocks.
- ❖ Because of slow cooling, they are coarsely crystallized, and their mineral crystals are large enough to be seen with the naked eye.
- ❖ Examples: Granite, Syenite, and Gabbro are typical plutonic igneous rocks.





Igneous Rocks - Volcanic

- ❖ These rocks are formed by the cooling and solidification of molten lava erupted onto the Earth's surface during volcanic eruptions.
- ❖ As the lava cools rapidly, the rocks develop fine-grained or glassy textures because crystals do not have sufficient time to grow.
- ❖ They are generally hard, dense, and durable, making them suitable for construction purposes.
- ❖ Examples of such rocks are basalt, trap and rheolite.





Igneous Rocks - Hypabyssal

- ❖ The rocks formed on account of cooling and solidification of molten magma at a shallower depth of about 2 to 3 km below the surface of earth.
- ❖ The rate of cooling is moderate, resulting in medium-grained rocks with crystals that are partly coarse and partly fine.
- ❖ They possess properties intermediate between plutonic and volcanic rocks.
- ❖ Examples of such rocks are aplites, dolerites, etc.





Igneous Rocks - Examples





Sedimentary Rocks

- ❖ Sedimentary rocks, derived from the word "sediment" meaning small particles, are the most abundant rock type blanketing much of Earth's surface.
- ❖ These rocks originate from the breakdown of existing rocks into tiny fragments through natural forces such as wind, flowing water, glaciers, and atmospheric gases.
- ❖ Over time, these particles accumulate and compact to form new sedimentary rock layers.





Sedimentary Rocks

- ❖ Weathering and erosion are natural forces that break down rocks into smaller particles.
- ❖ These particles, or sediments, are carried by rivers, lakes, seas, and oceans, where they settle over millions of years.
- ❖ Over time, the accumulated sediments are compacted under their own weight, transforming into solid, dense rocks known as secondary rocks.
- ❖ Because these rocks originate from deposited sediments, they are aptly named sedimentary rocks.
- ❖ In India sedimentary rocks can be found in regions like Rajasthan, Madhya Pradesh, and Andhra Pradesh.



Sedimentary Rocks - Classification

Clastic Rocks

- ❖ Clastic sedimentary rocks originate from the accumulation and solidification of fragmented particles derived from older rocks. These sediments typically settle in environments such as river valleys, lakes, and ocean basins.
- ❖ Representing the most abundant category of sedimentary rocks, clastics include types like sandstones, shales, breccias, and conglomerates.
- ❖ Sandstones primarily consist of silica grains that have been tightly cemented or compacted through natural geological processes.
- ❖ Due to their durability and aesthetic appeal, sandstones are widely valued as excellent building materials.



Sedimentary Rocks - Classification

Chemically Derived Sedimentary Rocks

- ❖ Numerous sedimentary rocks originate through chemical processes, often forming when minerals precipitate out of water in rivers, lakes, and especially oceans due to evaporation and other factors.
- ❖ During weathering and erosion, elements from existing rocks dissolve into water, which can become saturated over time, leading to the deposition of these minerals.
- ❖ Over long periods, the accumulation of these mineral deposits and evaporites can create substantial rock formations.
- ❖ Examples of such chemically formed sedimentary rocks include limestone, gypsum, anhydrite, and rock salt.
- ❖ It is important to note that these rocks are generally not utilized as building materials.



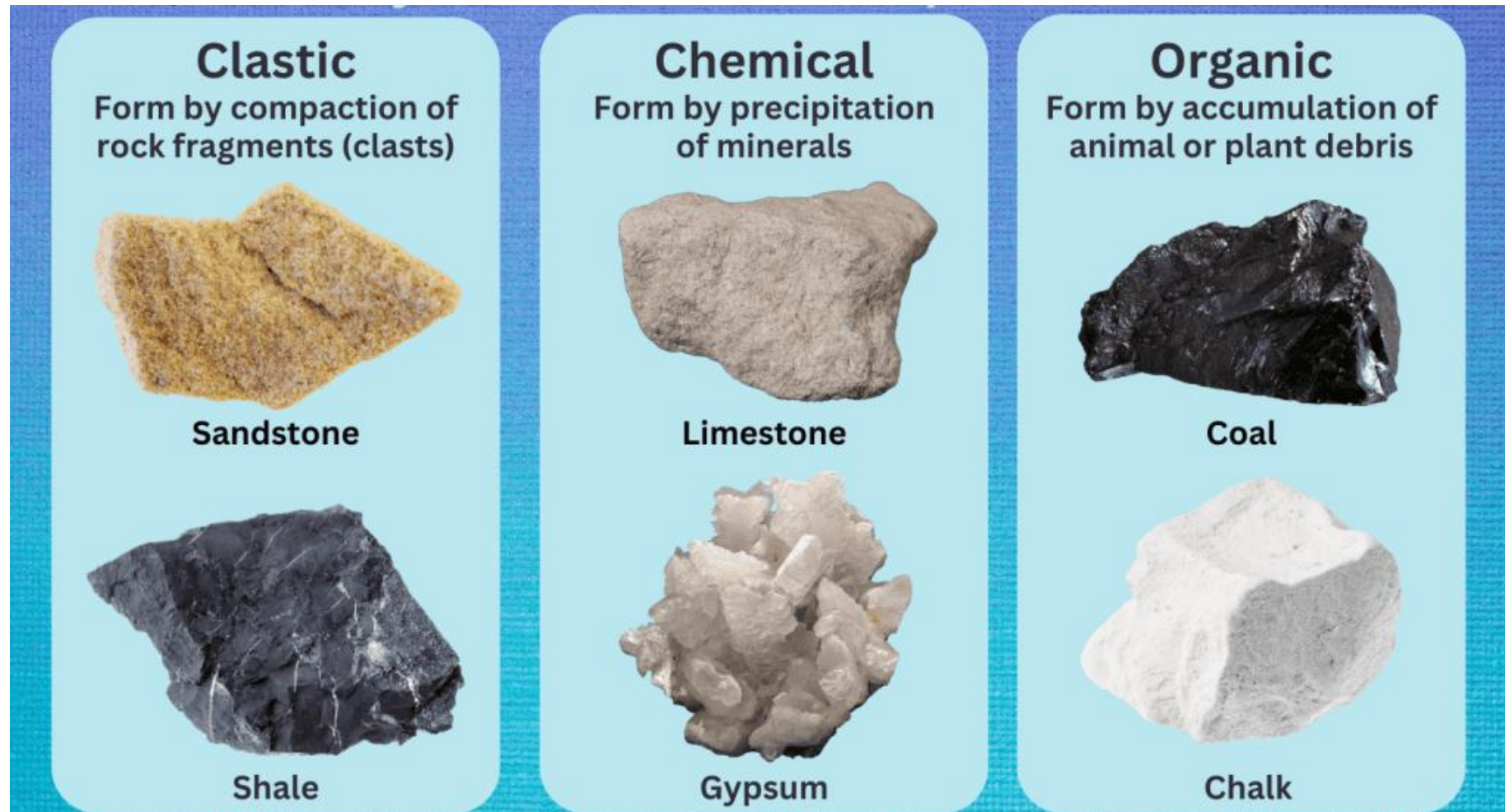
Sedimentary Rocks - Classification

Sedimentary Rocks Formed from Organic Material

- ❖ Over 70% of Earth's surface is covered by vast oceans and seas, teeming with diverse marine life.
- ❖ Many sea creatures possess hard skeletal parts composed primarily of calcium and magnesium carbonates.
- ❖ These skeletal remains accumulate on the ocean floor in specific areas, gradually building up thick layers.
- ❖ Over time, these deposits undergo compaction and cementation, transforming into sedimentary rocks formed from organic origins.
- ❖ Limestone is a prime example of such organically derived sedimentary rock, constituting a significant portion of mountain ranges worldwide, including the Himalayas.



Sedimentary Rocks - Classification





Metamorphic Rocks

- ❖ The term "metamorphic" comes from Greek roots meaning "change" (meta) and "form" (morph).
- ❖ These rocks originate from either igneous or sedimentary types.
- ❖ Metamorphism is the process where existing rocks transform due to elevated heat, pressure, and chemical conditions.
- ❖ When a rock is exposed to higher temperatures, altered stress conditions, and reactive fluids, it gradually changes its internal structure and chemical makeup.
- ❖ The rock that results from this transformation is known as a metamorphic rock.
- ❖ Two classic examples of metamorphic rocks are marble and slate.
- ❖ Marble originates when sedimentary limestone undergoes slow, natural heating, often due to magma rising beneath limestone formations.



Metamorphic Rocks - Classification





Silicious Rocks

❖ Silica is the predominant constituent of this rock and is more than 50% of the bulk composition of the rock. Some sedimentary and metamorphic rocks are entirely made of silica. These rocks are very strong and hence may be treated as good building stones. Examples of these rocks are granite, sandstone, gneiss.





Argillaceous Rocks

- ❖ Argillaceous rocks are sedimentary rocks mainly composed of clay and silt. They are formed by the deposition and compaction of fine-grained sediments over a long period.
- ❖ These rocks are soft, fine-textured, and have low strength, especially when wet. Common examples include **shale, claystone, and mudstone**.
- ❖ They are widely used in the manufacture of **bricks, tiles, cement, and ceramic products**, but are generally not suitable as load-bearing building stones due to their low strength and poor water resistance.





Calcareous Rocks

- ❖ Calcareous rocks are sedimentary rocks mainly composed of calcium carbonate (CaCO_3).
- ❖ They are formed from the accumulation of shells, corals, and other marine organisms or by the precipitation of calcium carbonate from water.
- ❖ These rocks are generally hard, durable, and easy to cut and shape. Common examples include limestone, chalk, and travertine.
- ❖ Calcareous rocks are widely used in the manufacture of cement and lime, as building stones, road construction materials, and decorative architectural works.





Stratified Rocks

- ❖ Stratified rocks are rocks that are formed in distinct horizontal layers or strata due to the gradual deposition of sediments over time.
- ❖ These layers may vary in thickness, colour, and composition. Stratified rocks can be easily split along the bedding planes.
- ❖ Common examples include sandstone, limestone, and shale. They are widely used as building stones, flooring materials, road aggregates, and in the manufacture of cement, depending on their properties





Unstratified Rocks

- ❖ Unstratified rocks are rocks that do not have distinct layers or strata.
- ❖ They are mainly formed by the cooling and solidification of molten magma, resulting in a uniform and compact structure.
- ❖ These rocks are generally hard, strong, and durable. Common examples include granite, basalt, and trap.
- ❖ Due to their high strength and weather resistance, they are widely used in building construction, road works, bridges, dams, and railway ballast.





Foliated Rocks

- ❖ Foliated rocks are metamorphic rocks that have a layered or banded structure due to the alignment of minerals under high pressure and temperature.
- ❖ These rocks can be split easily along their layers. They are generally strong, durable, and have an attractive appearance.
- ❖ Common examples include slate, schist, and gneiss.
- ❖ Foliated rocks are widely used for roofing, flooring, wall cladding, paving, and decorative building works.





Quarrying of Stones

- ❖ Stones occur in nature in the form of natural rock masses forming hills and valleys.
- ❖ They have to be broken or extracted from the available masses to be then used for different purposes.
- ❖ The process of extracting stones from natural rock deposits near the earth's surface is called quarrying.
- ❖ Quarrying is different than mining used for extraction of ores for metal in that in mining we go deep below the surface, the top surface is left intact, and the ores are extracted from under the earth. In quarrying, however, the process of extraction starts from the surface.



Selection of Quarrying Site

- ❖ The quarry site should have good quality stone that is strong, durable, and free from cracks and defects.
- ❖ It should have large and sufficient stone deposits to ensure economical extraction.
- ❖ The site should be easily accessible by road or rail for transportation of stones.
- ❖ The quarry should be located near the construction site to reduce transportation costs.
- ❖ Labour, water, electricity, and machinery should be readily available.
- ❖ The ground should have proper drainage to prevent water accumulation during quarrying.
- ❖ The site should be safe from landslides and flooding and should provide stable working conditions.
- ❖ The quarry should cause minimum environmental impact and comply with government regulations.
- ❖ There should be sufficient space for storage, processing, and movement of equipment.
- ❖ The overall cost of extraction, processing, and transportation should be economical.



Reflection Spot

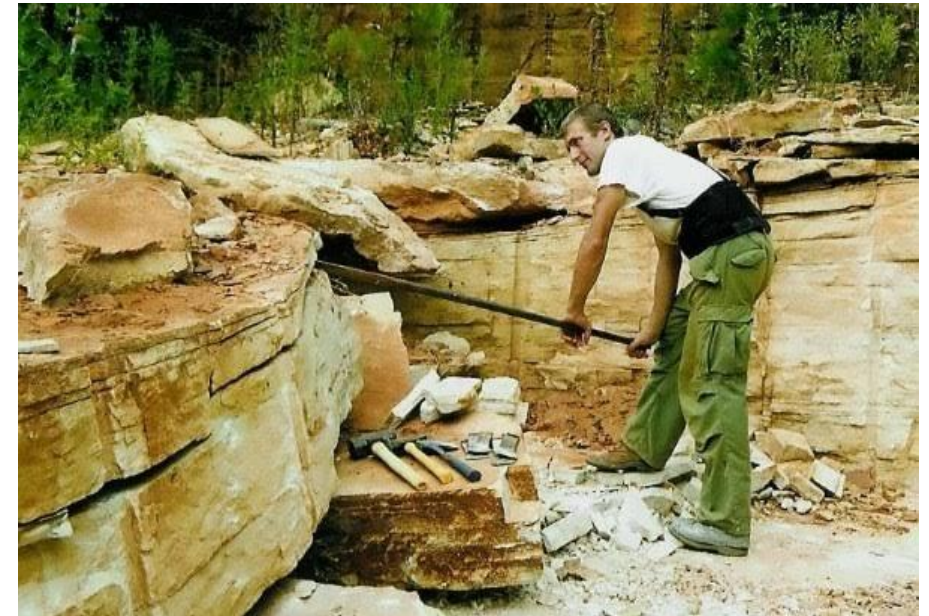
- ❖ Which is the biggest stone quarry site in India?
- ❖ **Answer: Jodhpur Sandstone Quarries**
- ❖ Which state in india has highest number of stone quarry sites?
- ❖ **Answer: Rajasthan**
- ❖ Which stone was used in the construction of the Red Fort?
- ❖ **Answer: Red sandstone was primarily used in the construction of the Red Fort in Delhi.**
- ❖ Which explosive is commonly used in stone quarry blasting?
- ❖ **Answer: Modern quarries commonly use ANFO (Ammonium Nitrate Fuel Oil) or emulsion explosives under controlled blasting procedures.**
- ❖ Which is the largest marble market in Asia?
- ❖ **Answer: Kishangarh, Rajasthan is the largest marble market and processing hub in Asia.**



Methods of Quarrying

Digging:

- ☐ Used for extracting soft stones available near the ground surface.
- ☐ Suitable for laterite, soft limestone, and weathered sandstone.
- ☐ Topsoil and loose material are removed before quarrying.
- ☐ Stones are dug out using picks, shovels, crowbars, and spades.
- ☐ No explosives or heavy machinery are required.
- ☐ Simple and economical method.
- ☐ Suitable for small-scale quarrying.
- ☐ Slow process with low production.
- ☐ Not suitable for hard rocks.





Methods of Quarrying

Heating:

- ☐ Used for quarrying hard rocks such as granite and basalt.
- ☐ Fire is lit on the rock surface to heat it.
- ☐ Cold water is poured on the heated rock.
- ☐ Sudden cooling creates cracks due to thermal shock.
- ☐ Cracked stones are removed manually.
- ☐ No explosives are required.
- ☐ Simple traditional method.
- ☐ Slow and labour-intensive process.
- ☐ Rarely used today because of high fuel consumption and environmental concerns.





Methods of Quarrying

Wedging:

- ☐ Used for extracting medium and hard stones.
- ☐ Holes are drilled in a straight line on the rock surface.
- ☐ Steel wedges and shims are inserted into the holes.
- ☐ Hammer blows split the rock along the required line.
- ☐ Produces uniform stone blocks.
- ☐ Causes minimum damage to the stone.
- ☐ No blasting vibrations or noise.
- ☐ Time-consuming and labour-intensive.
- ☐ Commonly used for granite, marble, and sandstone.





Methods of Quarrying

Blasting:

- ☐ Most common method for large-scale quarrying.
- ☐ Suitable for hard rocks such as granite, basalt, and limestone.
- ☐ Blast holes are drilled in the rock.
- ☐ Explosives and detonators are placed in the holes.
- ☐ Controlled blasting breaks the rock into pieces.
- ☐ Broken stones are collected and transported.
- ☐ Fast and economical method.
- ☐ Produces high output in less time.
- ☐ Causes noise, dust, and ground vibrations & requires strict safety precautions.





Methods of Quarrying

Channeling:

- ☐ Modern method used for extracting dimension stones.
- ☐ Suitable for marble, sandstone, and limestone.
- ☐ Channeling machines cut vertical and horizontal grooves.
- ☐ Large rectangular stone blocks are separated from the rock.
- ☐ Produces smooth and accurate cuts & reduces wastage of stone.
- ☐ Improves productivity and quality.
- ☐ Requires skilled operators and heavy machinery.
- ☐ High initial installation and maintenance cost.
- ☐ Widely used in commercial stone quarries.





Dressing of Stones

- ❖ The Dressing of stone is defined as “The process of giving a proper size, shape and finish to the roughly broken stones as obtained from the quarry.”
- ❖ This is done with the help of hand tools like a pickaxe, chisels, etc., or with the help of machines.
- ❖ This process is done manually or mechanically. A dressed stone is fit for use in a particular situation in a building.





Objectives of Stone Dressing

- (a) To reduce the size of the big blocks of stones so that they are converted to easily lift-able pieces. This reduction in size is generally carried out at the quarry itself because that saves a lot of transportation costs.
- (b) To give a proper shape to the stone: It is known that stones can be used at different places in the building, e.g., in foundations, in walls, in arches, or for flooring, each situation will require a proper shape.
- (c) To obtain an appealing finish: In a residential building, stones are used not only because of their extra strength, hardness, and durability but also because of their aesthetic value.

Methods of Stone Dressing





Selection of Stones for Construction

1. The stone should have high compressive strength to safely carry structural loads.
2. It should be hard and tough to resist wear, abrasion, and impact.
3. The stone should be durable and able to withstand weathering and environmental conditions.
4. It should have low water absorption to prevent dampness, frost damage, and deterioration.
5. The stone should be free from cracks, cavities, veins, and other defects.
6. It should have a uniform colour and texture for a good appearance.
7. The stone should be resistant to fire, chemicals, and atmospheric effects.
8. The stone should be economical, considering quarrying, transportation, and maintenance costs.
9. The selection of stone should be based on the type of structure and its intended application (e.g., foundation, masonry, flooring, paving, or decorative work).

THANK
YOU